Week 5: Continuous telemetry can replace a 48–72 hour lab lag

Weekly evidence artifact · Week 5 · Trimester 1, 2026

1. Theory anchor

Process analysis and bottleneck identification [10].

2. Artifact

	Today	Proposed
Sampling	Manual rounds across 20,000+ wellheads	Continuous SCADA downhole telemetry
Delay	48–72 h lab assay [6]	Near real time
Escalation	—	Pressure above 12 bar alerts Environmental Safety Director [6]
Fallback	—	Drift above 3% → 6-hour manual hydrometer rounds

3. Company A vs. Company B insight

Cameco shows process control paying off in production [3]; Kazatomprom's lag causes late valve triage and extra reagent.

4. Transfer verdict

Pilot 90 days on one Inkai wellfield block.

5. Evidence + verification

Target: 5%–6% less acid per kg U. This is a **proposed pilot target** benchmarked on IAEA/Cameco material [3] [6], not a verified Kazatomprom result.

6. Mini-recommendation

The Inkai Operations Manager should run the 90-day pilot; progress signal: acid use per kg U against baseline.

7. Learning progression & AI disclosure

I learned a 48–72 hour delay is a cost, not an inconvenience. AI helped structure the before/after table; thresholds are my proposals.

Companies: Cameco (A) and NAC Kazatomprom JSC (B). Source numbers [n] refer to the Evidence page of the ePortfolio.
Proposed targets and thresholds are design assumptions and pilot targets, not verified Kazatomprom data.